

Italian_exam_A002_ORD25-English

Italian Maturità — Esame di Stato, Mathematics (A002)

Ordinary session 2025 — Second written test Source: *Ministero dell'istruzione e del merito*

Valid for the Scientific Lyceum (indirizzi LI02, LI03, LI15, LI1S, LI22, LI23, LI31, LI32, LIA2, LIAO, LIB2, LIC2, LID2, LII2, LII3, LII4, LIIS, LIS2, EA02, EA10).

Instructions: The candidate must solve **one** of the two problems and answer **4** of the questions in the questionnaire.

Problem 1

“Reason is nothing without imagination.” — Descartes

Given ($r > 0$) and ($k < 0$), consider the circle (C_r) of centre the origin and radius (r), and the function $[f_k(x) = k|x|]$.

(a) Verify that (f_k) is continuous but not differentiable at ($x=0$) for every value of (k). Find the two values of (r) for which (C_r) and the graph of (f_k), for suitable (k), delimit a circular sector in the half-plane (y) of area (π) and perimeter (4π). Given that ($r=2$) is the larger such value, in the same Cartesian system (Oxy), sketch the circle (C_2) and the graph of (f_{-1}).

(b) Study the function $[g(x) = \dots]$ specifying its domain, symmetries, points of non-differentiability, intervals of monotonicity, and range. Verify that the graph of (g) coincides with the part of (C_2) in the half-plane (y). Explain why (g) is not invertible on its domain, and give the interval $[a; b]$ of maximum width, with ($b > 0$), on which (g) admits an inverse (h). What is the analytic expression of (h)?

(c) Let (A) be a point on the graph of (g) in the first quadrant, and let (M) and (R) be its orthogonal projections on the coordinate axes. Find the coordinates of (A) so that the quadrilateral ($AMOR$) has maximum area. After verifying that this quadrilateral is a square, show that it is also the one with maximum perimeter.

(d) Consider the function $[F(x) = \int_{-2}^x \sqrt{4-t^2} dt, x \in [-2, 2]]$. Determine ($F(2)$) and sketch the graph of (F) after studying its monotonicity and concavity. Write the equation of the tangent line to the graph of (F) at its point of inflection.

Problem 2

“Beauty is the right blending, in just proportions, of the finite and the infinite.” — attributed to Plato

The graphs ($_1$) and ($_2$) represent, respectively, the functions (f) and (g), defined on ($\mathbb{R}$), with analytic expressions $[f(x) = p(x)e^{p(x)}, g(x) = q(x)e^{p(x)}]$, where ($p(x)$) and ($q(x)$) are second-degree polynomials.

(a) Determine the polynomials ($p(x)$) and ($q(x)$) using the information from the figures, given that ($=$) is the abscissa of a stationary point of (f), and that ($-$), the abscissa of point (A), is a zero of (g).

(b) Setting ($p(x) = x - x^2$), study the function (f) by specifying the equation of the asymptote and the abscissas of the stationary and inflection points. Verify that the line ($x =$) is an axis of symmetry for ($_1$). Determine the range of (f), and indicate, as the real parameter (k) varies, the number of solutions of the equation ($f(x) = k$).

- (c) Setting $q(x)=1-x-x^2$, verify that $(1/)$ is the other zero of (g) and that triangle (ABC) is right-angled. Show that $(_1)$ and $(_2)$ have a unique point of intersection, whose coordinates are required. Consider points (P_1) and (P_2) on $(_1)$ and $(_2)$ respectively, with equal abscissa (x) ; compute the maximum length of the segment (P_1P_2) .
- (d) Compute the area of the bounded region (R) between $(_1)$, $(_2)$, and the (y) -axis. Then find the value of (t) for which the line $(x=t)$ delimits with the two graphs a region (R') of area equal to that of (R) .

Questionnaire (Quesiti)

- Given a triangle (ABC) , let (M) be the midpoint of side (BC) , and let (B') and (C') be points on (AB) and (AC) respectively, with $(AB'=AB)$ and $(AC'=AC)$. Prove that if the segments (MB') and (MC') are congruent, then so are the sides (AB) and (AC) .
- Consider the sphere $((x-1)^2+(y-2)^2+z^2=1)$ and the plane $(x-2y-2z+d=0)$. As the real parameter (d) varies, discuss whether $()$ is secant, tangent, or external to the sphere. Determine the value of (d) for which $()$ divides the sphere into two equal parts.
- Boccioni's Futurist work "*Unique Forms of Continuity in Space*" (1913), shown on the 20-cent coin, depicts a man advancing rapidly through space. Part of the profile highlighted in the figure, in a suitable reference frame, can be approximated by the function $[f(x)=$

$$\begin{cases} -4x^2 - 8x, & -1 \leq x \leq 0 \\ 1 + \tan\left(x + \frac{3}{4}\pi\right), & 0 < x \leq 2 \end{cases}$$

- $] \text{ Sketch the graph of } (f) \text{ after analysing its continuity and differentiability on } ([-1;2]).$
- Given a function (g) , differentiable on $()$ and such that $(g(/4)=g'(/4)=2)$, determine the equation of the normal line to the curve $(y=g(x)^2 x)$ at its point of abscissa $(/4)$.
 - Determine the value of the real parameter (k) such that the two curves $(y=e^x)$ and $(y=6-k \cdot e^{-x})$ are tangent to each other, and find the coordinates of the point of tangency.
 - Write a polynomial function (f) such that the line $(y=2x+3)$ is tangent to the graph of (f) at its point of abscissa 0 , and $[\int_0^3 f(x) dx = 9.]$
 - From Cicero, *De divinatione*, II, 21, 48 — about the "Venus throw," which consists of rolling 4 dice (4-sided) and obtaining 4 different outcomes. Assuming each face is equally likely, determine:
 - the probability of obtaining the Venus throw with 4 dice;
 - the probability of obtaining 4 equal numbers.
 - How many anagrams (with or without meaning) of the word **STUDIARE** exist? In how many of these can the word **ARTE** be read consecutively (as in "SUARTEDI")? How many anagrams of the word **VACANZA** exist?

"Mathematics knows no races or geographic boundaries; for mathematics, the cultural world is a single country." — D. Hilbert

Maximum duration: 6 hours. Scientific or graphing calculators are allowed, provided they have no symbolic algebraic processing capability and no internet connection.